In-Class Activity: IV Infusions

Module 3 - Intravenous Infusions · In-class activity · No date

Name _____ Date _____

Question 1 You are asked to recommend an IV infusion rate to achieve a steady-state concentration of 20 mg/L. The half-life of the drug is 5 hours and the volume of distribution is 16 L. What rate do you recommend?

Question 1 You are asked to recommend an IV loading dose that will rapidly achieve a steady-state concentration of 20 mg/L. The half-life of the drug is 5 hours and the volume of distribution is 16 L. What loading dose do you recommend?

Question 2 A drug with a half-life of 5 hours and a volume of distribution of 16 L is administered at 44.35 mg/hr WITH a 320 mg loading dose. What is the expected concentration of drug 3 hours after the start of the infusion?

Question 3 A drug with a half-life of 5 hours and a volume of distribution of 16 L is infused at 44.35 mg/hr WITHOUT a loading dose for 3 hours. What is the expected concentration of drug at the end of the infusion?

Question 4 A drug with a half-life of 5 hours and a volume of distribution of 16 L is infused at 44.35 mg/hr WITHOUT a loading dose for 3 hours. What is the expected concentration of drug 4 hours after cessation of the infusion?

5a A patient received a medication as a 500 mg IV loading dose and a simultaneous IV infusion of 75 mg/hr for 3 days. The half-life of the medication was 4 hours and the volume of distribution was 12 L. What is the expected steady-state concentration of the infusion?

5b The same patient is given a 750 mg loading dose rather than 500 mg, simultaneously with the same 75 mg/hr infusion. The half-life is 4 hours and the volume of distribution is 12 L. What is the steady-state concentration?

5c A medication is infused at 75 mg/hr, giving a steady-state concentration of 36.1 mg/L. Its half-life is 4 hours and its volume of distribution is 12 L. What is an appropriate loading dose to achieve that steady-state concentration immediately?

5d A medication with a half-life of 4 hours is infused at a constant rate with no loading dose. How long would it take to reach 90% of the desired steady-state concentration?

5e The same medication, with a half-life of 4 hours and a volume of distribution of 12 L, is infused at 150 mg/hr rather than 75 mg/hr. What steady-state concentration is expected?